RESTAURANT AND FOOD INSPECTIONS ANALYSIS

Final Project Report Documentation

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1. INTRODUCTION

Food safety inspections are essential to ensure restaurants and food establishments follow hygiene andpublic health regulations. This project analyzes restaurant inspection data to identify violations, inspectiontrends, risk levels, and business compliance patterns using MS Excel and Power BI.

The project focuses on transforming raw inspection datasets into meaningful dashboards and reports thathelp stakeholders monitor restaurant performance and food safety compliance.

2. PROJECT OBJECTIVE

- The main objectives of this project are:
- To analyze restaurant inspection records.
- To identify restaurants with repeated violations.
- To monitor food safety compliance levels.
- To classify inspection risk categories.
- To track inspection trends over time.
- To create interactive dashboards for decision-making.
- To improve reporting using data visualization techniques.

3. PROBLEM STATEMENT

Food inspection datasets usually contain large volumes of raw and inconsistent data with missing values,duplicate records, incorrect formats, and multiple violation categories. Manual analysis becomes difficult and time-consuming.

This project aims to clean, transform, model, and visualize restaurant inspection data to generate actionable insights regarding:

- High-risk restaurants
- Frequent violation categories
- Failed inspections
- Geographic inspection trends
- Compliance performance over time.

4. DATASET DESCRIPTION

The dataset contains restaurant and food inspection records collected from public food safety inspection databases.

Dataset feature:

Column Name	Description
Business Name	Name of the restaurant
business ID number	Unique ID number
Inspection Date	Date of inspection
City,location	Address and location
Business _postal_code	Location code
Business state	State of inspection
Inspection Type	Routine / Follow-up inspection
Risk Category	Low / Medium / High risk
Violation Code	Code assigned for violations
Violation Description	Details of food safety issue
Inspection Score	Score received during inspection
Inspection Result	Pass / Fail

5.tools and technologies used:

Tool	Purpose
MS Excel	Data cleaning and preprocessing
Power Query	Data transformation
Power BI	Dashboard creation and visualization
DAX	Measures and calculated columns

6. DATA CLEANING PROCESS

The raw dataset contained:

- Missing values
- Duplicate records
- Uneven date formats
- Incorrect text formatting
- Blank inspection scores
- Inconsistent risk categories

Cleaning Steps Performed in Excel and Power Query

- Removed duplicate rows.
- Handled missing values using Replace and Fill methods.
- Standardized date formats.
- Converted data types correctly.
- Split and merged columns where necessary.
- Standardized restaurant names and city names.
- Removed null and irrelevant records.

- Created separate tables for modeling.

Relationship type:

- One to one
- Many to one
- One to many
- Many to many

Dax measure:

Total restuarants
Total Inspections = COUNT('Fact_Inspections'[Inspection ID])

Total violations
Total Violations = COUNT('Fact_Inspections'[Violation Code])

Average inspection score

AverageScore = AVERAGE('Fact_Inspections'[Inspection Score])

Count of inspection type

Count of inspection type=count(fact inspections[inspection type])

Inspection _status

Inspection_status=if(fact inspections[inspection score]>=85,"good","needs improvement"])

Pending violation count

Pending violation count=count(fact inspection[violation count])

Year of inspection conducted

Year=year(fact inspection[inspection date])

Charts:

Stacked bar chart
Total restaurants by inspection date

AREA CHART
Total violations by business name

AREA CHART
Count of inspection date by inspection status

PIE CHART
Count of food inspection by inspection status
Food quality inspection pass rate is higher
*minimum got average percentage
*fail rate is less

KPI/GUAGE VISUALISATION
KPI CARD-Sum of inspection score and total restuarants by Inspection status.
GUAGE CARD-Total restaurants.

COLUMN CHART VISUALISATION
Total violations by city wise and risk status
*Comparing violations by cities have higher rate in good condition
*comparing violations in medium condition needs to be improved

TABLE /MATRIX
TABLE -Inspection Date wise analysis
*Total Restuarants-939
*Total Violations-939
*Pending violations count-329
*Risk status on particular inspection date.
MATRIX
***Analysing business name by year
*Sum of inspection score by particular year.

CLUSTERED COLUMN CHART
*Total restaurants and count of inspection by risk category.

10. DASHBOARD FEATURES
The Power BI dashboard includes:
Interactive filtering options
Dynamic KPI tracking
Risk-based restaurant analysis
Violation trend analysis
Geographic visualization
Responsive report design

11. KEY INSIGHTS

High-risk restaurants showed repeated hygiene violations.
Certain cuisine categories had more frequent inspection failures.
Violation counts increased during specific months.
Restaurants with lower inspection scores had higher repeat violations.
Some cities recorded higher numbers of food safety issues.